

Strategic Under-Specification: A Stackelberg Game Between User and Assistant

Pranjal Agarwal
BITS Pilani, Hyderabad Campus

Abstract

Users of coding assistants routinely write terse prompts, relying on the assistant to handle missing details through guessing or clarifying questions. We develop a contract-theoretic model of endogenous clarification in human–AI interaction. Prompt shortening is not driven by moral hazard: by the envelope theorem, improving raw guessing accuracy weakly improves user welfare despite shortening prompts, reflecting rational effort substitution ($m^*(a) \leq m^*(0)$). When clarification policies are opaque, naive users who overestimate clarification willingness ($a^\dagger > a_A$) under-specify relative to the assistant’s true policy ($m_{\text{naive}}^* < m_{\text{soph}}^*$), incurring quadratic dead-weight losses. Under Stackelberg commitment with transparent policies, however, an assistant with under-asking friction bias ($\lambda_A > c_Q$) suppresses clarification, which paradoxically forces sophisticated users to over-specify upstream ($m^* > m^{FB}$) to defensively insure against guessing errors.

Closing the loop under uniform *pooling* and effort-conditioned *screening* governance regimes, we establish that on the physically feasible domain ($m \leq k$), asking weakly dominates guessing whenever clarification is cost-effective ($c_Q < \Lambda$). Under pooling, provider asking friction ($\lambda_A > c_Q$) suppresses clarification ($\partial a^{SE}/\partial \lambda_A \leq 0$). This induces an equilibrium capability reversal (the “guessing trap”): higher raw guessing accuracy reduces the return to asking, leading biased assistants to slash clarification so sharply that users must write longer prompts ($d\mathbb{E}[m^*]/dg > 0$) and suffer lower welfare ($dW/dg < 0$). Under screening, provider bias induces adverse-selection downward distortion on terse users ($a_H^{SB} < a_H^B$) when low-cost effort saturates ($\kappa_L \leq \bar{\kappa}_L$), while boundary relaxation ($\bar{\kappa}_L < \kappa_L \leq c_Q^{eff}/k$) tempers this distortion. We delineate genuine screening distortion from technological clarification shutdown ($c_Q^{eff} \geq \Lambda$), show that nonlinear probability curvature generates interior trade-offs, and propose an empirical randomized controlled trial.

Keywords: Human–AI Interaction, Clarification Policies, Stackelberg Games, Contract Theory, Adverse Selection.

Code Availability: Simulation suite and verification scripts are available at <https://github.com/pranjal-0802/strategic-underspecification>.

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1 Introduction

A user typing a coding request faces a fundamental economic trade-off: spend cognitive effort articulating edge cases, naming conventions, and constraints, or leave them implicit and trust the assistant to infer them. The assistant, in turn, must decide for each unspecified detail whether to guess or to ask. Neither choice is made in a vacuum: the user’s specification effort depends on

what they expect the assistant to do with silence, and the assistant’s ask/guess policy is chosen (explicitly, or as an emergent property of training and product incentives) with some implicit model of how users will respond.

This paper provides a mechanism-design theory of endogenous clarification in AI assistance. Its central premise is that prompt shortening reflects rational effort substitution in response to automated clarification ($m^*(a) \leq m^*(0)$), while welfare-reducing under-specification arises when users naively anticipate more clarification than the assistant provides ($m_{\text{naive}}^* < m_{\text{soph}}^*$). When the assistant’s policy is transparently chosen as a Stackelberg leader, provider asking friction ($\lambda_A > c_Q$) suppresses clarification, forcing users to over-specify relative to first-best ($m^* > m^{FB}$) and generating an equilibrium capability trap where better guessing can perversely lengthen prompts and lower welfare. Section 2 begins with a decision-theoretic benchmark where the assistant’s policy is exogenous, proving that improved guessing accuracy alone weakly improves user welfare despite shortening prompts. Section 3 formalizes effort substitution and belief-driven under-specification under opaque policies. Sections 4–7 close the loop by modeling the assistant as a Stackelberg leader anticipating user best responses under pooling and screening instruments. Section 8 translates the equilibrium characterizations into falsifiable behavioral predictions. Section 10 evaluates exact nonlinear robustness and discusses institutional extensions.

1.1 Related literature

This work bridges contract theory and the emerging literature on formal models of human–AI interaction.

Contract theory and screening without transfers. Our screening framework builds on classical adverse-selection and mechanism design models (Mirrlees, 1971; Myerson, 1979, 1981; Baron and Myerson, 1982; Mussa and Rosen, 1978; Laffont and Tirole, 1986) that exploit the Spence–Mirrlees single-crossing property (Spence, 1973) to separate heterogeneous agents. However, unlike classical monopoly or regulatory screening where the principal uses monetary transfers to extract information rents, here the assistant screens strictly through an interactive service margin (the ask rate a offered in response to specification effort m). This connects to non-monetary screening and costly verification (Townsend, 1979; Ben-Porath et al., 2014), where information revelation is governed by operational friction rather than cash transfers.

Delegation, authority, and information design. Our analysis relates closely to the theory of delegation under asymmetric information (Holmström, 1984; Dessein, 2002), where a principal determines how much discretion to grant an agent, and to models of formal versus real authority (Aghion and Tirole, 1997), where an agent’s upstream effort is shaped by who retains authority over downstream implementation. In our setting, leaving attributes unspecified implicitly delegates their resolution to the assistant. Furthermore, because the assistant’s asking policy determines which state dimensions are actively resolved, our framework shares conceptual foundations with Bayesian persuasion and strategic information design (Kamenica and Gentzkow, 2011), while focusing on the effort-substitution trade-off in prompt articulation.

Costly communication and strategic disclosure. The user’s specification decision formalizes costly disclosure under strategic interpretation, connecting to strategic information transmission (Crawford and Sobel, 1982), voluntary quality disclosure (Grossman, 1981), and rational inattention (Sims, 2003). In our setting, under-specification arises not from an incentive to deceive, but from the user’s rational economizing on specification costs when anticipating automated assistance.

Assistance games and human–AI interaction. Our bi-level formulation models the interaction as a Stackelberg game (von Stackelberg, 1934), connecting conceptually to the literature on assistance games and cooperative inverse reinforcement learning (Hadfield-Menell et al., 2016), as well as incomplete contracting in AI alignment (Hadfield-Menell and Hadfield, 2019), where an autonomous assistant acts to maximize human utility under incomplete specification. However, while assistance games formalize the assistant’s uncertainty over the human’s underlying *objective or reward function*, in our setting the economic task objective is known and shared (downstream success value V and error loss L). Instead, the informational asymmetry resides entirely in the user’s private *specification cost* κ . The strategic margin in our model is therefore not reward learning or preference alignment, but rather the contract-theoretic trade-off between prompt specification effort and automated clarification under costly communication. Our model provides a contract-theoretic microfoundation for this ask-versus-guess margin under asymmetric specification costs.

Clarification and interactive NLP. A parallel empirical literature in natural language processing and information retrieval investigates algorithms for generating and ranking clarifying questions in conversational search and dialog (Rao and Daumé III, 2018; Aliannejadi et al., 2019; Zamani et al., 2020). Recent empirical evaluations show that state-of-the-art large language models often exhibit conversational fragility by guessing prematurely under ambiguity rather than asking clarifying questions (Laban et al., 2025). In the computer science literature, dialog policies are typically evaluated against fixed query benchmarks or trained via reinforcement learning against static conversational corpora. However, these approaches treat user inputs as exogenous queries rather than the endogenous choice of an optimizing agent who strategically anticipates the assistant’s clarification policy. Our framework provides the contract-theoretic foundation that this engineering literature lacks, formally endogenizing prompt specification effort as an equilibrium best response to the assistant’s asking policy.

2 Environment and the Decision-Theoretic Benchmark

2.1 Setup

A task has k binary, independent, intent-relevant attributes $\theta = (\theta_1, \dots, \theta_k) \in \{0, 1\}^k$, each drawn i.i.d. uniform and privately known to the user. The user chooses a specification effort level $m \in \{0, \dots, k\}$ and truthfully reveals m attributes.¹ Specifying m attributes costs $c(m)$, with $c(0) = 0$, $c' \geq 0$, $c'' \geq 0$.

For each of the $k - m$ unspecified attributes, the assistant either *asks* (revealing the true value, at a cost to the user) or *guesses*, succeeding independently with probability $g \in (0, 1)$. We model the assistant’s policy at this stage as a uniform ask rate $a \in [0, 1]$ applied to every unspecified attribute, so each is correctly resolved with probability $q(a, g) = a + (1 - a)g$. Task success requires *all* k attributes to be correctly resolved (conjunctive: one wrong parameter or convention typically breaks the deliverable). The user’s payoff is

$$U(m; a, g) = V q(a, g)^{k-m} - c(m) - a c_Q (k - m),$$

with $V > 0$ the value of success and $c_Q > 0$ the user’s cognitive cost of answering one clarifying question. We treat m as continuous on $[0, k]$ for tractability.

¹We assume truthful revelation conditional on the choice to specify: the strategic margin is *how much* to say, not whether to lie about what is said. A cheap-talk layer on top of this costly-specification layer is a natural extension; see Section 10.

2.2 Proposition 1: accuracy alone does not create moral hazard

Proposition 1. *Fix a . Let $U^*(g) = \max_{m \in [0, k]} U(m; a, g)$. Then U^* is weakly increasing in g .*

Proof. For any fixed $m \in [0, k]$, the user's objective $U(m; a, g) = Vq(a, g)^{k-m} - c(m) - ac_Q(k - m)$ is pointwise non-decreasing in g because $\partial U / \partial g = V(k - m)q(a, g)^{k-m-1}(1 - a) \geq 0$. As the upper envelope of a family of weakly increasing functions, the maximized utility $U^*(g) = \max_{m \in [0, k]} U(m; a, g)$ is weakly increasing in g directly, without requiring differentiability. Under differentiability, the envelope theorem confirms that at the interior optimum $m^*(g)$,

$$\frac{dU^*}{dg} = \left. \frac{\partial U}{\partial g} \right|_{m=m^*(g)} = V(k - m^*)q^{k-m^*-1}(1 - a) \geq 0.$$

The envelope theorem lets us ignore dm^*/dg : because m^* is chosen optimally, its first-order response to g has no first-order effect on U^* . $\square$

Assumption 1 (Regularity). *In the relevant range of m , $(k - m) \ln q(a, g) \geq -1$.*

Remark 1 (Interpreting Assumption 1 and Model Concavity). *The regularity condition $(k - m) \ln q(a, g) \geq -1$ provides a sufficient condition for decreasing differences in (m, g) , which guarantees monotone comparative statics via Topkis's theorem (Topkis, 1978, 1998; Milgrom and Shannon, 1994). The direction is pinned down by the cross-partial of utility: for fixed a ,*

$$\frac{\partial^2 U}{\partial m \partial g} = -V(1 - a)q^{k-m-1}[(k - m) \ln q + 1].$$

Thus U exhibits decreasing differences in (m, g) whenever $(k - m) \ln q + 1 \geq 0$, which is precisely Assumption 1. Because this condition is most demanding at $m = 0$, it reduces on the primitive domain to $k \ln q \geq -1$. Since $\ln q \approx -(1 - q)$ for q near 1, this condition requires $k(1 - q) \lesssim 1$: the expected number of residual unresolved attributes under guessing is at most about one. If this condition fails ($k \ln q < -1$), U locally exhibits increasing differences, and higher guessing accuracy can locally induce higher specification effort before falling as $q \rightarrow 1$.

We also note that in the exact conjunctive model, the gross benefit Vq^{k-m} is strictly convex in m (since $\partial^2[Vq^{k-m}]/\partial m^2 = Vq^{k-m}(\ln q)^2 > 0$). Global concavity of U in m therefore requires the specification cost curvature $c''(m) = \kappa$ to strictly exceed $Vq^{k-m}(\ln q)^2$ across $[0, k]$. Under the linear-risk specification introduced in Section 4, by contrast, gross benefit is linear in m , guaranteeing that U is globally strictly concave in m for all $\kappa > 0$.

Corollary 1. *Under Assumption 1, $m^*(g)$ is weakly decreasing in g (decreasing differences in (m, g) ; Topkis's theorem).*

Proof. Immediate from the cross-partial computed in Remark 1: Assumption 1 is exactly the condition $\partial^2 U / \partial m \partial g \leq 0$ on the relevant range, which is decreasing differences in (m, g) ; Topkis's theorem then gives $m^*(g)$ weakly decreasing. $\square$

Thus, holding the assistant's policy fixed, users specify less as the assistant becomes more adept at guessing, yet they are never worse off for doing so. Proposition 1 serves primarily as a baseline: it applies the envelope theorem to a single agent optimizing against a technology parameter that enters the payoff with a non-negative derivative, a property that holds generally in models of this class. Its purpose is not to present a surprising result, but to establish the rigorous benchmark against which claimed moral hazard must be evaluated, ensuring that Proposition 2 isolates the true source of welfare loss. This frames the central question of this paper: what happens once the policy a is no longer exogenous, but chosen by a strategic assistant that anticipates the user's response?

3 Sources of Strategic Under-Specification

Proposition 1 assumed the assistant’s ask rate is fixed and that users correctly anticipate $q(a, g)$. Both assumptions can fail.

3.1 Cost externality

If a third party (such as a code reviewer or downstream production system) bears part of the cost of an erroneous guess, the user’s private parameters (κ, V) understate the social costs, causing the user to under-specify relative to the social optimum at any fixed g , an inefficiency that worsens as g rises by Corollary 1. While this constitutes a standard Pigouvian externality, our primary focus lies in strategic and informational friction.

3.2 Miscalibrated belief updating under silent failure

If incorrect guesses are not always detected, and the detection probability $d(m)$ is increasing in m , users infer g from experience rather than observing it directly. Under-specifying users are consequently less likely to catch their own errors, generating an upward-biased estimate $\hat{g}$ that reinforces under-specification. This mechanism represents a dynamic learning process rather than a static equilibrium outcome; we formalize its testable implications in Section 8 (prediction P5).

3.3 Belief about the assistant’s policy and equilibrium under-specification

When the clarification policy a is endogenous, the user’s specification decision depends on their belief regarding the assistant’s asking rate. Let users be *sophisticated* (correctly inferring the assistant’s true operational ask rate a_A) or *naive* (believing the assistant follows a higher clarification rate $a^\dagger > a_A$, such as the unconstrained welfare-optimal policy).

Definition 1 (Effort Substitution and Belief-Driven Under-Specification). *A user interacting with an automated assistant modifies upstream specification effort m across two distinct economic margins:*

- (i) **Rational effort substitution:** $m^*(a) \leq m^*(0)$, where the user economizes on prompt articulation by delegating attribute resolution to automated assistance; and
- (ii) **Belief-driven under-specification:** $m^*(\hat{a}) < m^*(a)$, where a user holding optimistic or miscalibrated beliefs $\hat{a} > a$ under-specifies relative to the assistant’s true operational ask rate.

In human–AI interaction, varying the clarification rate a engages two opposing economic channels:

1. *The accuracy channel:* A higher ask rate increases effective resolution accuracy $q(a, g) = a + (1 - a)g$. Under decreasing differences (Assumption 1), greater accuracy reduces the marginal value of upstream prompt effort, pushing specification downward.
2. *The question-cost channel:* Each clarifying question imposes an interactive answering cost c_Q on the user. Because the total answering cost $ac_Q(k - m)$ is decreasing in specification m , a higher ask rate increases the marginal penalty of leaving attributes unspecified, pushing specification upward (pre-emption).

Under the linear-risk specification analyzed in Sections 4–6, where the net benefit of clarification $s \equiv \Lambda - c_Q > 0$ is positive, the cross-partial derivative of user utility is:

$$\frac{\partial^2 U}{\partial m \partial a} = -(\Lambda - c_Q) < 0.$$

The accuracy channel strictly dominates the question-cost channel, ensuring that U exhibits strictly decreasing differences in (m, a) .

Proposition 2 (Prompt Under-Specification under Naive Beliefs). *Under the linear-risk specification with $\Lambda > c_Q$, naive users choose weakly lower specification effort than sophisticated users:*

$$m_{naive}^* \leq m_{soph}^*,$$

holding strictly ($m_{naive}^* < m_{soph}^*$) for all interior types satisfying $\kappa > \frac{\Lambda(1-a^\dagger)+a^\dagger c_Q}{k}$. Consequently, naive users achieve weakly lower true welfare, $U(m_{naive}^*; a_A) \leq U(m_{soph}^*; a_A)$, strictly so on the interior domain. Holding raw guessing accuracy g fixed, naive users under-specify relative to the assistant's true operational policy and suffer an equilibrium welfare loss.

Proof. Under linear risk on the physical domain $m \in [0, k]$, the follower's best response to an anticipated ask rate a is $m^*(\kappa; a) = \min\{k, \beta(\kappa; a)\}$, where $\beta(\kappa; a) \equiv \frac{\Lambda(1-a)+ac_Q}{\kappa}$. Since $\Lambda > c_Q$, the unconstrained response $\beta(\kappa; a)$ is strictly decreasing in a : $\partial\beta/\partial a = (c_Q - \Lambda)/\kappa < 0$. A naive user anticipating $a^\dagger > a_A$ therefore satisfies $\beta(\kappa; a^\dagger) < \beta(\kappa; a_A)$. For low-cost types with $\kappa \leq \frac{\Lambda(1-a^\dagger)+a^\dagger c_Q}{k}$, we have $\beta(\kappa; a^\dagger) \geq k$, so $\beta(\kappa; a_A) > k$ as well. Both beliefs therefore yield boundary saturation: $m_{naive}^* = m_{soph}^* = k$, and true welfare is identical ($U(m_{naive}^*; a_A) = U(m_{soph}^*; a_A)$). For all interior types satisfying $\kappa > \frac{\Lambda(1-a^\dagger)+a^\dagger c_Q}{k}$, we have $\beta(\kappa; a^\dagger) < k$, which guarantees:

$$m_{naive}^* = \beta(\kappa; a^\dagger) < \min\{k, \beta(\kappa; a_A)\} = m_{soph}^*.$$

Because true utility $U(m; a_A)$ is strictly concave in m with unique global maximizer on $[0, k]$ at m_{soph}^* , any choice $m \neq m_{soph}^*$ incurs a strict utility loss. On the fully interior domain where $m_{soph}^* = \beta(\kappa; a_A) < k$, evaluating the quadratic shortfall directly yields:

$$U(m_{soph}^*; a_A) - U(m_{naive}^*; a_A) = \frac{\kappa}{2} (m_{soph}^* - m_{naive}^*)^2 = \frac{(\Lambda - c_Q)^2}{2\kappa} (a^\dagger - a_A)^2 > 0,$$

confirming that interior naive users are strictly worse off.

Notice that on the interior domain, the under-specification gap is:

$$m_{soph}^* - m_{naive}^* = \frac{(\Lambda(g) - c_Q)(a^\dagger - a_A)}{\kappa} = \frac{(L(1-g) - c_Q)(a^\dagger - a_A)}{\kappa}.$$

Differentiating with respect to raw guessing accuracy g yields:

$$\frac{\partial(m_{soph}^* - m_{naive}^*)}{\partial g} = -\frac{L(a^\dagger - a_A)}{\kappa} < 0.$$

As the assistant's autonomous guessing capability improves, the distortion gap between naive and sophisticated users strictly contracts, because guessing errors become less frequent and the economic penalty of misplaced trust declines. $\square$

Remark 2 (Exact Model Counterexample to Monotone Under-Specification). *In the exact conjunctive model $U(m, a) = Vq(a, g)^{k-m} - c(m) - ac_Q(k - m)$, the cross-partial derivative is:*

$$\frac{\partial^2 U}{\partial m \partial a} = -V(1 - g)q^{k-m-1}[(k - m) \ln q + 1] + c_Q.$$

Even when Assumption 1 holds, the question-answering cost term $+c_Q$ can dominate if questions are costly relative to baseline task value V . For example, with $k = 5, V = 10, \kappa = 0.5, g = 0.8, c_Q = 2, a_A = 0.1$, and $a^\dagger = 0.6$, Assumption 1 holds across the domain, yet sophisticated users choose $m_{soph}^ = 3.15$ while naive users choose $m_{naive}^* = 3.92$. Here, the naive user anticipates substantial clarification costs and over-specifies upstream to avoid conversational interruptions. Proposition 2 thus isolates the regime where clarification is socially cost-effective ($\Lambda > c_Q$), which is the relevant environment for helpful automated assistants.*

This identifies the mechanism of welfare loss under opaque policies: an unobserved gap between the user’s belief regarding the assistant’s policy and the assistant’s actual policy. In Section 4, we close the loop by modeling the assistant as a strategic Stackelberg leader. Under transparent commitment, sophisticated users know the true policy a_A . As we show in Sections 5–6, provider asking friction ($\lambda_A > c_Q$) suppresses clarification ($a < 1$), leading sophisticated users to over-specify relative to first-best ($m^*(a) > m^{FB}$) to defensively insure against guessing errors.

4 The Assistant’s Policy as a Stackelberg Game

4.1 General formulation

The assistant (leader) commits ex ante to a policy π before meeting any user. A population of users (followers) has privately known specification-cost type $\kappa \sim F$ on $[\kappa_L, \kappa_H]$, with cost $c(m; \kappa) = \kappa m^2/2$, and best-responds $m^*(\kappa; \pi) = \arg \max_m U(m; \pi, \kappa)$. The leader’s own objective Π need not equal U : we write

$$\Pi_\kappa(m, a) = \mu_A U(m, a; \kappa) - (\lambda_A - c_Q)a(k - m),$$

where $\mu_A \in (0, 1]$ discounts the leader’s weight on true downstream user welfare, c_Q is the user’s true cognitive cost of answering one clarifying question, and λ_A is the leader’s own perceived or institutional cost per clarifying question. In commercial AI systems, misalignment along these margins arises from concrete engineering and economic forces:

1. *Asking friction bias* ($\lambda_A > c_Q$): Reinforcement learning from human feedback (RLHF) and preference optimization algorithms (e.g., DPO) commonly evaluate conversation turns in isolation. Reward models systematically penalize “evasive” clarifying questions in favor of immediate code generation because users in rating pools favor direct answers. Furthermore, multi-turn interactions increase token costs, context-window expansion, and inference latency, creating provider-side pressure to suppress clarification turns.
2. *Downstream welfare discounting* ($\mu_A < 1$): Turn-level user satisfaction signals (such as prompt acceptance or immediate positive feedback) capture immediate session fluency. In contrast, the true economic loss L from broken conventions, subtle edge-case errors, or security defects is typically discovered hours or days later in downstream execution, test suites, or code review. Consequently, training objectives tuned to immediate session retention tend to discount downstream correctness.

When asking friction is aligned ($\lambda_A = c_Q$), the leader's objective satisfies $\Pi_\kappa \equiv \mu_A U(\cdot; \kappa)$, sharing the identical preferences and policy maximizers as true user welfare for any positive welfare weight $\mu_A > 0$. Downstream welfare discounting $\mu_A < 1$ creates policy distortion only through interaction with excess asking friction ($\lambda_A \neq c_Q$), which inflates the effective asking friction per unit of altruism $(\lambda_A - c_Q)/\mu_A$. The leader solves $\pi^* = \arg \max_\pi \mathbb{E}_{\kappa \sim F}[\Pi_\kappa(m^*(\kappa; \pi), \pi)]$, a bilevel program.

Remark 3 (Existence). *For Model I below ($\pi = a \in [0, 1]$, a compact scalar policy), $m^*(\kappa; a)$ is continuous in a (strict concavity of U in m plus the implicit function theorem), so $\mathbb{E}_\kappa[\Pi_\kappa]$ is continuous in a and a maximizer exists on $[0, 1]$ by Weierstrass's theorem. For Model II below (a finite menu over two types), the feasible set is a non-empty, compact subset of $\mathbb{R}^4$ cut out by continuous (bilinear in (m, a) and quadratic in m) IC/IR constraints on the compact domain $[0, k]^2 \times [0, 1]^2$. The constraint set is non-empty because the degenerate pooling allocation $(m^*(a), a, m^*(a), a)$ satisfies all constraints. By Weierstrass's theorem, the continuous leader objective attains a global maximum; no ironing or ex-ante randomization is needed with only two types.*

For tractability in what follows we replace the exact conjunctive success probability q^{k-m} with its first-order (small-risk) linear approximation around $q \rightarrow 1$: expected loss $\approx L(1 - q)(k - m)$. When linearizing total task value $Vq^{k-m} \approx V - V(1 - q)(k - m)$, the per-attribute loss parameter satisfies $L = V$; more generally, $L \leq V$ parameterizes settings where individual unresolved attributes cause localized defect losses without necessarily destroying the entire project value. With $\Lambda \equiv L(1 - g)$, the user's payoff becomes

$$U(m, a; \kappa) = V - \Lambda(1 - a)(k - m) - \frac{\kappa}{2}m^2 - a c_Q (k - m).$$

Throughout Sections 4.2–6, we maintain the physical feasibility constraint $m \in [0, k]$, so that followers cannot specify more attributes than exist in the task. When unconstrained first-order conditions yield $m^*(\kappa; a) = \frac{\Lambda(1-a)+ac_Q}{\kappa} > k$ (which occurs for low-cost types with $\kappa < \Lambda/k$), best-response effort saturates at the physical boundary $m^* = k$.

4.2 The first-best benchmark and physical feasibility

Before analyzing either strategic regime, we characterize the fully-informed, fully-aligned social optimum:

$$(m^{FB}(\kappa), a^{FB}(\kappa)) = \arg \max_{m \in [0, k], a \in [0, 1]} U(m, a; \kappa).$$

Proposition 3 (First-Best Policy on the Feasible Domain). *On the physically feasible domain $m \in [0, k]$, under the linear-risk specification with $\Lambda > c_Q$:*

- (i) *Asking weakly dominates guessing for every cost type $\kappa > 0$: $a^{FB}(\kappa) = 1$, with optimal specification $m^{FB}(\kappa) = \min\{k, c_Q/\kappa\}$.*
- (ii) *For any type with $\kappa > c_Q/k$, asking strictly dominates guessing: $U(m^*(1), 1) > U(m^*(0), 0)$. For boundary types with $\kappa \leq c_Q/k$, specification effort saturates at $m^{FB} = k$, leaving zero residual attributes ($k - m = 0$) and rendering the asking policy payoff-irrelevant.*

Proof. For any fixed specification effort $m \in [0, k]$, the partial derivative of utility with respect to a is:

$$\frac{\partial U}{\partial a} = (\Lambda - c_Q)(k - m).$$

Because $\Lambda > c_Q$, this derivative is strictly positive for all $m < k$ and zero at $m = k$. Now consider the joint optimization over $m \in [0, k]$ and $a \in \{0, 1\}$. Under $a = 1$, the user chooses

$m^*(1) = \min\{k, c_Q/\kappa\}$. Under $a = 0$, the user chooses $m^*(0) = \min\{k, \Lambda/\kappa\}$. We evaluate the three type intervals:

- If $\kappa \geq \Lambda/k$, both best responses remain interior: $m^*(1) = c_Q/\kappa \leq m^*(0) = \Lambda/\kappa \leq k$. Evaluating the welfare difference:

$$U(m^*(1), 1; \kappa) - U(m^*(0), 0; \kappa) = (\Lambda - c_Q) \left(k - \frac{\Lambda + c_Q}{2\kappa} \right).$$

Since $\kappa \geq \Lambda/k > \frac{\Lambda + c_Q}{2k}$, the bracketed term satisfies $k - \frac{\Lambda + c_Q}{2\kappa} > k - k = 0$. Hence $U(m^*(1), 1; \kappa) > U(m^*(0), 0; \kappa)$ strictly.

- If $c_Q/k < \kappa < \Lambda/k$, unconstrained effort under $a = 0$ exceeds k , so specification under guessing saturates at the boundary $m^*(0) = k$, yielding $U(k, 0; \kappa) = V - \frac{\kappa}{2}k^2$. Under asking, effort remains interior at $m^*(1) = c_Q/\kappa < k$, yielding $U(m^*(1), 1; \kappa) = V - c_Qk + \frac{c_Q^2}{2\kappa}$. Subtracting:

$$U(m^*(1), 1; \kappa) - U(k, 0; \kappa) = \frac{\kappa k^2}{2} - c_Qk + \frac{c_Q^2}{2\kappa} = \frac{(\kappa k - c_Q)^2}{2\kappa} > 0.$$

- If $\kappa \leq c_Q/k$, specification effort saturates at $m^* = k$ under both policies, so $U(k, 1; \kappa) = U(k, 0; \kappa) = V - \frac{\kappa}{2}k^2$.

Therefore, $a^{FB}(\kappa) = 1$ is globally optimal across all types $\kappa > 0$, with indifference only on the boundary set $\kappa \leq c_Q/k$ where $m^{FB} = k$. $\square$

Remark 4 (The Unclipped Threshold κ^* and Genuine Asking–Guessing Trade-offs). *In unconstrained algebra without the feasibility constraint $m \leq k$, setting the difference $(\Lambda - c_Q)(k - \frac{\Lambda + c_Q}{2\kappa})$ to zero yields the formal threshold $\kappa^* \equiv \frac{\Lambda + c_Q}{2k}$. However, since $\Lambda > c_Q$, we have $\kappa^* < \Lambda/k$ strictly. The region $\kappa < \kappa^*$ where guessing formally appeared to beat asking in unclipped algebra lies entirely in the unphysical domain where unconstrained effort Λ/κ exceeds k (the user would specify more attributes than exist in the task). Once $m \leq k$ is enforced, asking weakly dominates guessing everywhere under linear risk because asking resolves uncertainty at lower unit cost ($c_Q < \Lambda$) while allowing the user to economize on specification effort.*

Genuine interior thresholds where $a^{FB} \in \{0, 1\}$ strictly partitions the type space arise from two natural richer specifications:

1. Exact conjunctive success: Under $Vq(a, g)^{k-m} - c(m) - ac_Q(k - m)$, the marginal gain from clarification is $(k - m)[V(1 - g)q^{k-m-1} - c_Q]$. When baseline accuracy g is high or $k - m$ is small, this marginal gain turns negative, making guessing strictly optimal for low-cost types who specify most attributes upstream.
2. Heterogeneous attribute stakes: If attributes vary in importance with stakes Λ_i , efficiency prescribes asking only on high-stakes attributes ($\Lambda_i > c_Q$) and guessing on low-stakes attributes ($\Lambda_i < c_Q$).

5 Model I: A Pooling (Scalar) Policy

The leader is restricted to a single population-wide ask rate $a \in [0, 1]$ and cannot condition on user-specific attributes or observed effort. This provides a natural benchmark for policies embedded uniformly at training time or applied without user-level profiling.

5.1 User best response

The FOC $\partial U / \partial m = \Lambda(1 - a) - \kappa m + a c_Q = 0$ gives, with $\beta(\kappa) = \Lambda / \kappa$ and $\gamma(\kappa) = (\Lambda - c_Q) / \kappa$,

$$m^*(\kappa; a) = \beta(\kappa) - a \gamma(\kappa), \quad \frac{\partial m^*}{\partial a} = -\gamma(\kappa) = \frac{c_Q - \Lambda}{\kappa}.$$

We assume interior specification levels $m^* \in (0, k)$; for low-cost types where unconstrained $\beta(\kappa) \geq k$, specification effort saturates at $m^* = k$. The closed-form population moments $\bar{R}_0$ and $\bar{\gamma}$ below are exact when best responses remain interior, and serve as tractable linear approximations when boundary clipping binds; as verified in Section 10 and our computational robustness suite, the qualitative regime structure and corner-solution properties are strictly preserved under boundary clipping and exact payoffs alike.

Remark 5 (Non-monotone effect of ask rate on specification). *Unlike Corollary 1, the sign of $\partial m^* / \partial a$ here does not depend on type: it is positive for every type iff $c_Q > \Lambda$, i.e., iff answering a question costs users more than the expected loss it avoids. When that condition holds, a higher population ask rate induces users to specify more, not less; they pre-empt costly clarifying questions by articulating the answer directly. This introduces a new economic force relative to Section 2, where a operated solely through effective accuracy, and generates a distinct, directly testable prediction (prediction P1' below).*

5.2 The leader's problem

Let $R(a, \kappa) = k - m^*(\kappa; a)$ denote the residual unspecified attributes. In the pooling regime, the assistant commits ex ante to an operational ask rate $a \in [0, 1]$ applied population-wide, without observing or directly contracting on individual specification effort.

Evaluating the general leader objective from Section 4.2 at the follower's optimal best response $m^*(\kappa; a) = \beta(\kappa) - a \gamma(\kappa)$:

$$\Pi_\kappa(m^*(\kappa; a), a) = \mu_A U(m^*(\kappa; a), a; \kappa) - (\lambda_A - c_Q) a (k - m^*(\kappa; a)).$$

Substituting $U(m^*(\kappa; a), a; \kappa) = V - \Lambda(1 - a)(k - m^*) - \frac{\kappa}{2}(m^*)^2 - a c_Q (k - m^*)$, using $k - m^*(\kappa; a) = R_0(\kappa) + a \gamma(\kappa)$, and defining the effective resolution stake $s \equiv \Lambda - c_Q$ and the assistant's excess asking friction $b \equiv \lambda_A - c_Q$, expanding in powers of a yields:

$$\Pi_\kappa(m^*(\kappa; a), a) = \mu_A \left(V - \Lambda k + \frac{\Lambda^2}{2\kappa} \right) + (\mu_A s - b) R_0(\kappa) a + \frac{(\mu_A s - 2b) \gamma(\kappa)}{2} a^2.$$

Averaging over $\kappa \sim F$ requires no moments beyond the population sufficient statistics already defined, $\bar{R}_0 = k - \mathbb{E}[\beta]$ and $\bar{\gamma} = \mathbb{E}[\gamma] = s \mathbb{E}[1/\kappa]$:

$$\Pi(a) = \text{const} + \alpha_1 a + \alpha_2 a^2,$$

where $\text{const} = \mu_A(V - \Lambda k) + \frac{\mu_A \Lambda^2}{2} \mathbb{E}[1/\kappa]$, and

$$\alpha_1 \equiv (\mu_A s - b) \bar{R}_0, \quad \alpha_2 \equiv \frac{(\mu_A s - 2b) \bar{\gamma}}{2}.$$

Lemma 1 (Leader Objective as User Welfare with an Asking Friction Wedge). *For any allocation (m, a) , the leader's objective satisfies:*

$$\Pi(m, a) = \mu_A \left[U(m, a) - \tilde{\rho} a (k - m) \right],$$

where $\tilde{\rho} \equiv \frac{\lambda_A - c_Q}{\mu_A}$ is the normalized excess asking friction. The leader's objective incorporates true user welfare but introduces a per-question friction wedge $\tilde{\rho}$. In the leader's evaluation, clarifying questions are priced at an effective cost $c_Q^{eff} \equiv c_Q + \tilde{\rho}$. However, this is not an equivalence to a benevolent social planner, because followers continue to best-respond against their true private question cost c_Q , not c_Q^{eff} . Because $\mu_A > 0$ enters strictly as an overall positive scalar multiplier on the bracketed term, optimal leader policies depend on (μ_A, λ_A) solely through the excess friction ratio $\tilde{\rho}$.

Proposition 4 (Stackelberg-optimal pooling rate). *Provided $\alpha_2 < 0$ (the second-order condition for strict concavity, equivalent to $2(\lambda_A - c_Q) > \mu_A(\Lambda - c_Q)$), the leader's unconstrained stationary point is*

$$a^{SE} = -\frac{\alpha_1}{2\alpha_2} = -\frac{(\mu_A s - b) \bar{R}_0}{(\mu_A s - 2b) \bar{\gamma}}.$$

When this stationary point lies in $[0, 1]$, it is the unique global maximum of $\Pi(a)$ on $[0, 1]$.

Corollary 2 (Friction bias suppresses the pooling ask rate on the interior branch). *On the interior branch $\alpha_2 < 0$, differentiating the closed form in Proposition 4 directly with respect to λ_A (holding primitives $\mu_A, \Lambda, c_Q, \bar{R}_0, \bar{\gamma}$ fixed) gives:*

$$\frac{\partial a^{SE}}{\partial \lambda_A} = -\frac{\bar{R}_0}{\bar{\gamma}} \frac{\mu_A s}{(\mu_A s - 2b)^2} = -\frac{\bar{R}_0}{\bar{\gamma}} \frac{\mu_A(\Lambda - c_Q)}{[\mu_A(\Lambda - c_Q) - 2(\lambda_A - c_Q)]^2}.$$

For populations supported on the interior specification domain ($\kappa \geq \Lambda/k$ for all types in the support), follower effort satisfies $m^*(\kappa; 0) = \Lambda/\kappa \leq k$, ensuring that expected residual unspecified attributes under silence satisfy $\bar{R}_0 = k - \mathbb{E}[\beta] > 0$. Because $\bar{\gamma} > 0$, $\mu_A > 0$, and $s > 0$, it follows that:

$$\frac{\partial a^{SE}}{\partial \lambda_A} \leq 0.$$

On this interior domain, higher perceived clarification friction λ_A strictly suppresses the assistant's equilibrium asking rate. When the population includes lower-cost types ($\kappa < \Lambda/k$) whose effort saturates at $m^* = k$, the unclipped linear moment decomposition is modified by boundary clipping; numerical optimization across the clipped distribution confirms that higher friction λ_A continues to weakly suppress asking, transitioning monotonically from $a^{SE} = 1$ to $a^{SE} = 0$.

Corollary 3 (Continuous transition of the pooling ask rate across friction regimes). *Under asking friction bias ($\lambda_A > c_Q$), the second-order coefficient satisfies $\alpha_2 < 0$ if and only if $\tilde{\rho} > s/2$. On this concave branch, the optimal pooling policy transitions continuously on $[0, 1]$ across three distinct friction regimes:*

1. Mild bias ($\tilde{\rho} \leq \tilde{\rho}_1$): For $\tilde{\rho} \leq \tilde{\rho}_1 \equiv \frac{s(\bar{R}_0 + \bar{\gamma})}{\bar{R}_0 + 2\bar{\gamma}}$, the unconstrained stationary point satisfies $-\alpha_1/(2\alpha_2) \geq 1$, so the constrained maximum on $[0, 1]$ is universal clarification: $a^{SE} = 1$.
2. Moderate bias ($\tilde{\rho}_1 < \tilde{\rho} < s$): The unconstrained stationary point is strictly interior, so the optimal ask rate is uniquely:

$$a^{SE} = -\frac{\alpha_1}{2\alpha_2} = \frac{(s - \tilde{\rho}) \bar{R}_0}{(2\tilde{\rho} - s) \bar{\gamma}} \in (0, 1),$$

which is strictly decreasing in friction ($\partial a^{SE}/\partial \tilde{\rho} < 0$).

3. Prohibitive friction ($\tilde{\rho} \geq s$, equivalently $c_Q^{eff} \geq \Lambda$): The leader unconstrainedly prefers zero clarification ($-\alpha_1/(2\alpha_2) \leq 0$), shutting down asking completely: $a^{SE} = 0$.

The endpoint crossover threshold $\tilde{\rho}^* \equiv \frac{s(\bar{R}_0 + \bar{\gamma}/2)}{\bar{R}_0 + \bar{\gamma}}$ satisfies $\tilde{\rho}_1 < \tilde{\rho}^* < s$. When $\alpha_2 \geq 0$ (weak convexity, as in the unbiased case $\tilde{\rho} = 0 \leq s/2$), $\Pi(a)$ is convex and $\tilde{\rho}^*$ uniquely dictates the bang-bang corner choice. When $\alpha_2 < 0$, $\Pi(a)$ is strictly concave and the maximum at $\tilde{\rho} = \tilde{\rho}^*$ is interior ($a^{SE} \in (0, 1)$), with $\Pi(1) = \Pi(0)$ indicating payoff equality at the endpoints rather than a discontinuous jump.

Corollary 4 (Pooling is a corner solution in the unbiased case). *In the unbiased case ($\mu_A = 1, \lambda_A = c_Q \implies b = 0$), the quadratic coefficient is:*

$$\alpha_2 = \frac{s\bar{\gamma}}{2} = \frac{s^2 \mathbb{E}[1/\kappa]}{2} = \frac{(\Lambda - c_Q)^2 \mathbb{E}[1/\kappa]}{2} \geq 0 \quad \text{always.}$$

The second-order condition for an interior maximum therefore fails: $\Pi(a)$ is weakly convex on $[0, 1]$, its stationary point is a local minimum, and the true optimum is always a corner, $a^{SE} \in \{0, 1\}$, not an interior compromise. Comparing the two corners directly in unclipped algebra:

$$\Pi(1) - \Pi(0) = \alpha_1 + \alpha_2 = s \left(\bar{R}_0 + \frac{\bar{\gamma}}{2} \right) = (\Lambda - c_Q) \left[k - \frac{\Lambda + c_Q}{2} \mathbb{E} \left[\frac{1}{\kappa} \right] \right],$$

which yields the formal unclipped threshold $k^* \equiv \frac{\Lambda + c_Q}{2} \mathbb{E}[1/\kappa]$. When follower best responses are physically constrained to the feasible domain $m \leq k$, Proposition 3 implies that asking ($a = 1$) weakly dominates guessing for every individual type in the support, so unbiased pooling under physical feasibility selects $a^{SE} = 1$ unconditionally. Genuine interior trade-offs between asking and guessing at the population level emerge under the exact conjunctive success model, as analyzed in Section 10.

Proposition 5 (Equilibrium Reversal of Guessing Capability: The Guessing Trap). *Under asking friction bias ($\lambda_A > c_Q$), an increase in assistant guessing capability g affects the interior pooling clarification rate according to:*

$$\frac{da^{SE}}{dg} < 0 \iff \mu_A \bar{R}_0(g) > (\mu_A s(g) - b) \mathbb{E} \left[\frac{1}{\kappa} \right] (1 - 2a^{SE}).$$

In particular:

- (i) If $a^{SE} \geq 1/2$, the condition holds unconditionally, so $\frac{da^{SE}}{dg} < 0$. If $a^{SE} < 1/2$, the derivative can reverse when $\bar{R}_0$ is small.
- (ii) When $\frac{da^{SE}}{dg} < 0$, prompt lengthening and welfare decline are mathematically equivalent and hold simultaneously if and only if

$$-\frac{da^{SE}}{dg} > \frac{L(1 - a^{SE})}{\Lambda(g) - c_Q},$$

reversing the comparative statics of Proposition 1 in Stackelberg equilibrium:

- **Equilibrium prompt lengthening:** $\frac{d\mathbb{E}[m^*]}{dg} > 0$.
- **Equilibrium welfare decline:** $\frac{dW}{dg} < 0$.

Proof. On the interior branch, the stationary point is $a^{SE}(g) = -\frac{\alpha_1(g)}{2\alpha_2(g)} = \frac{(\mu_A s(g) - b)\bar{R}_0(g)}{(2b - \mu_A s(g))\bar{\gamma}(g)}$, where $\Lambda(g) = L(1-g)$, $s(g) = L(1-g) - c_Q$, $\bar{R}_0(g) = k - \Lambda(g)\mathbb{E}[1/\kappa]$, and $\bar{\gamma}(g) = s(g)\mathbb{E}[1/\kappa]$. Differentiating with respect to g and collecting terms in $\mu_A, b, s(g), \bar{R}_0(g)$, and $\mathbb{E}[1/\kappa]$ yields:

$$\frac{da^{SE}}{dg} = -\frac{L}{\bar{\gamma}(g)(2b - \mu_A s(g))} \left[\mu_A \bar{R}_0(g) - (\mu_A s(g) - b)\mathbb{E}\left[\frac{1}{\kappa}\right] (1 - 2a^{SE}) \right].$$

Because the prefactor $-\frac{L}{\bar{\gamma}(g)(2b - \mu_A s(g))}$ is strictly negative on the concave interior branch ($2b > \mu_A s$), it follows that $da^{SE}/dg < 0$ if and only if $\mu_A \bar{R}_0(g) > (\mu_A s(g) - b)\mathbb{E}[1/\kappa](1 - 2a^{SE})$. When $a^{SE} \geq 1/2$, $(1 - 2a^{SE}) \leq 0$, ensuring the condition is satisfied unconditionally. Conversely, when $a^{SE} < 1/2$, the right-hand side is strictly positive; for example, with $k = 12, L = 17, c_Q = 4.5, \lambda_A = 11.5$, and $\kappa \sim \text{Uniform}[1.4, 1.5]$, unclipped effort remains interior but a^{SE} increases from 0.211 to 0.230 as g rises from 0.05 to 0.10.

By the chain rule, total equilibrium follower effort satisfies:

$$\frac{dm^*}{dg} = \frac{\partial m^*}{\partial g} + \frac{\partial m^*}{\partial a} \frac{da^{SE}}{dg} = \frac{1}{\kappa} \left[-L(1 - a^{SE}) - s(g) \frac{da^{SE}}{dg} \right].$$

Similarly, by the envelope theorem on follower choice of m^* , total equilibrium user welfare $W(g) \equiv \mathbb{E}[U(m^*(g), a^{SE}(g); g)]$ satisfies:

$$\frac{dW}{dg} = \mathbb{E} \left[\frac{\partial U}{\partial g} \Big|_{m^*} \right] + \mathbb{E} \left[\frac{\partial U}{\partial a} \Big|_{m^*} \right] \frac{da^{SE}}{dg} = \mathbb{E}[(k - m^*)] \left[L(1 - a^{SE}) + s(g) \frac{da^{SE}}{dg} \right].$$

Because $\mathbb{E}[k - m^*] > 0$ and $\mathbb{E}[1/\kappa] > 0$, prompt lengthening ($d\mathbb{E}[m^*]/dg > 0$) and welfare decline ($dW/dg < 0$) are mathematically equivalent and hold simultaneously if and only if $-da^{SE}/dg > \frac{L(1 - a^{SE})}{s(g)}$.

Evaluating this mechanism on the physically admissible domain for binary attributes ($g \in [0.50, 0.65]$ under benchmark parameters $k = 10, L = 10, c_Q = 2, b = 1.2, \kappa \sim \text{Uniform}[0.2, 0.6]$): as guessing capability improves from $g = 0.50$ to $g = 0.60$, the equilibrium ask rate collapses from $a^{SE} = 0.661$ to $a^{SE} = 0.424$. In response to this withdrawal of clarification, average prompt specification expands from $\mathbb{E}[m] = 7.728$ to 7.953, and expected user welfare declines from $W = 82.134$ to 81.843 (verified in `underspec_sim/verifications/verify_prop5_guessing_trap.py`). $\square$

This property proves essential for the comparative analysis in Section 7: Model I exhibits welfare loss relative to the first best primarily through objective bias, which shifts a^{SE} (Corollary 2) and suppresses the operational clarification rate.

6 Model II: A Screening (Menu) Policy

Now let the leader condition its ask rate on the user's declared specification level (equivalently, by the revelation principle, offer a menu of bundles that users self-select into). For tractability and clean analytical exposition, we restrict attention to two types $\kappa_L < \kappa_H$ with population shares f_L, f_H (the continuum-type extension using the standard Mirrlees technique is discussed in Section 10). Let the leader choose bundles $(m_L, a_L), (m_H, a_H)$ subject to

$$\text{IC}_\kappa : U(m_\kappa, a_\kappa; \kappa) \geq U(m_{\kappa'}, a_{\kappa'}; \kappa), \quad \text{IR}_\kappa : U(m_\kappa, a_\kappa; \kappa) \geq \underline{U},$$

to maximize $f_L \Pi_{\kappa_L}(m_L, a_L) + f_H \Pi_{\kappa_H}(m_H, a_H)$. We normalize the follower's reservation utility to $\underline{U} = 0$, representing complete task abandonment without AI assistance. In practice, users also possess the endogenous outside option of proceeding with a non-asking baseline assistant ($a = 0$), which yields reservation utility $\underline{U}(\kappa) \equiv \max_m U(m, 0; \kappa)$. For low-cost types ($\kappa \leq \Lambda/k$), this baseline yields $\underline{U}(\kappa) = V - \frac{\kappa}{2}k^2$. Under the optimal screening menu derived below, type L receives the bundle $(k, 0)$, yielding $U(k, 0; \kappa_L) = V - \frac{\kappa_L}{2}k^2 \equiv \underline{U}(\kappa_L)$. Under this endogenous outside option, individual rationality binds exactly at the bottom of the type distribution (IR_L holds with equality), while type H 's rent $U_H - \underline{U}(\kappa_H) > 0$ remains strictly positive. This provides a natural counterpart to classical regulatory screening (such as Baron–Myerson), where the least-costly agent is driven to its participation boundary.

6.1 Single-crossing

The marginal rate of substitution between m and a along an indifference curve in (m, a) -space is given by the slope $\text{MRS}(m, a; \kappa) \equiv -\frac{\partial U / \partial m}{\partial U / \partial a}$. From user utility $U(m, a; \kappa) = V - \Lambda(1 - a)(k - m) - \frac{\kappa}{2}m^2 - ac_Q(k - m)$, we have $\partial U / \partial m = \Lambda(1 - a) + ac_Q - \kappa m$ and $\partial U / \partial a = s(k - m)$, where $s \equiv \Lambda - c_Q > 0$ is the net social gain from asking. Hence,

$$\text{MRS}(m, a; \kappa) = \frac{\kappa m - [\Lambda - as]}{s(k - m)}, \quad \frac{\partial \text{MRS}}{\partial \kappa} = \frac{m}{s(k - m)} > 0,$$

which holds strictly for all interior specification levels $0 < m < k$ under the maintained condition $s > 0$. Higher- κ types therefore have strictly steeper indifference curves in (m, a) -space: they require more compensating clarification service a per unit increase in specification effort m . This satisfies the standard Spence–Mirrlees single-crossing property (Spence, 1973) and underwrites the screening analysis below.

6.2 No distortion without bias

Proposition 6 (Efficiency survives private information when the leader is unbiased). *If $\mu_A = 1$ and $\lambda_A = c_Q$ (so $\Pi_\kappa \equiv U(\cdot; \kappa)$ for every κ), then implementing each type's own first-best bundle, $(m_\kappa, a_\kappa) = (m^{FB}(\kappa), a^{FB}(\kappa))$ from Proposition 3, is incentive compatible, and no further mechanism design can do better for the leader.*

Proof. $(m^{FB}(\kappa), a^{FB}(\kappa))$ maximizes $U(\cdot; \kappa)$ by definition, so for any $\kappa' \neq \kappa$,

$$U(m^{FB}(\kappa), a^{FB}(\kappa); \kappa) \geq U(m^{FB}(\kappa'), a^{FB}(\kappa'); \kappa)$$

holds trivially: no bundle can give type κ more utility than its own maximizer. IC holds with no distortion needed. Since $\Pi_\kappa \equiv U(\cdot; \kappa)$, the leader's payoff from serving each true type is maximized at that type's own first best, so this menu is also leader-optimal. $\square$

This result highlights a fundamental distinction between human–AI coordination and classical adverse selection (such as Baron–Myerson or Laffont–Tirole). In regulatory and auction models, the principal pays monetary transfers to extract surplus and thus treats informational rents as a pure loss. Here, by contrast, the leader's objective can coincide directly with user welfare; when it does, incentive compatibility is costless, and private information incurs no welfare penalty. Distortion in this setting is therefore a pure signature of objective misalignment, not of asymmetric information per se.

Remark 6 (Degeneracy of the Unbiased Menu on the Asking Margin). *Under Proposition 3, $a^{FB}(\kappa) = 1$ for every type in the population on the physically feasible domain $m \leq k$. The unbiased first-best menu is therefore $(m^{FB}(\kappa), 1) = (\min\{k, c_Q/\kappa\}, 1)$, which offers uniform asking $a = 1$ across all types and leaves followers to freely choose their preferred specification effort. This menu does not screen across the asking margin: scalar pooling at $a^{SE} = 1$ (Corollary 4) implements each type's first best identically. Consequently, under linear risk on the feasible domain, the welfare gap between pooling and screening at zero bias is identically zero ($\Pi_{II} - \Pi_I = 0$). A positive instrument-restriction gap at zero bias emerges only in settings where first best itself prescribes different asking policies across types, such as the exact conjunctive success model (Section 10) or under attribute loss heterogeneity.*

6.3 Distortion under bias

When $\Pi_\kappa \neq U(\cdot; \kappa)$ (bias present), the leader's preferred bundle per type,

$$(m_\kappa^B, a_\kappa^B) = \arg \max_{m,a} \Pi_\kappa(m, a)$$

(computable in closed form by the same method as Proposition 3, with Λ, c_Q replaced by their μ_A, λ_A -weighted counterparts), no longer automatically satisfies IC, because Π_κ and the user's true $U(\cdot; \kappa)$ now rank bundles differently.

Proposition 7 (Downward distortion of the high-cost type's service). *Assume $\Lambda > c_Q$, under-asking bias ($0 < \lambda_A - c_Q < \mu_A(\Lambda - c_Q)$, so that $c_Q < c_Q^{eff} < \Lambda$), cost heterogeneity $\kappa_L < c_Q^{eff}/k < \kappa_H$, and task complexity $k > k_{crit}$ (defined in (1)). Let population shares be $f_L, f_H > 0$ with $f_L + f_H = 1$, and define the relative share weight $t \equiv \frac{f_H}{f_L} \frac{s - \tilde{\rho}}{s}$. Then:*

1. **Incentive violation of unconstrained menu:** *If the low-cost type receives the unconstrained boundary bundle $(m_L^B, a_L^B) = (k, 0)$ and the high-cost type receives $(m_H^B, a_H^B) = (c_Q^{eff}/\kappa_H, 1)$, IC_L is strictly violated if and only if $\kappa_L(k + m_H^B) > 2c_Q$.*
2. **Downward distortion:** *Under the optimal incentive-compatible menu, the high-cost type's ask rate is strictly distorted downward relative to the leader's unconstrained preference:*

$$a_H^{SB} < a_H^B = 1,$$

in the saturation regime $\kappa_L \leq \bar{\kappa}_L$.

3. **Closed form under saturation of low-cost effort:** *If $\kappa_L \leq \bar{\kappa}_L \equiv \frac{c_Q^{eff} + tc_Q}{k(1+t)}$, the leader sets $(m_L^{SB}, a_L^{SB}) = (k, 0)$, and the high-cost contract is uniquely given in closed form by:*

$$m_H^* = \frac{\Lambda \tilde{\rho}}{s(\kappa_H - \kappa_L) + \tilde{\rho}\kappa_L}, \quad a_H^{SB} = 1 - \frac{\frac{\kappa_L(k + m_H^*)}{2} - c_Q}{s}.$$

In this saturation regime, downward distortion $1 - a_H^{SB}$ intensifies as κ_L increases toward $\bar{\kappa}_L$. At $m_L = k$, zero residual attributes remain ($k - m_L = 0$), so a_L is payoff-irrelevant to both parties.

4. **Boundary relaxation:** *If $\bar{\kappa}_L < \kappa_L < c_Q^{eff}/k$, the leader optimally offers type L clarification service ($a_L^{SB} = 1, m_L^{SB} < k$) to increase U_L , which relaxes IC_L and moderates the downward distortion on type H (a_H^{SB} rises toward 1 as $\kappa_L \rightarrow c_Q^{eff}/k$, and can reach $a_H^{SB} = 1$ when f_L is sufficiently small).*

Proof. We establish the result in five analytical steps:

Step 1: Unconstrained biased optima. The leader objective per type is $\Pi_\kappa(m, a) = \mu_A[V - \Lambda(1 - a)(k - m) - \frac{\kappa}{2}m^2 - ac_Q^{eff}(k - m)]$, where $c_Q^{eff} \equiv c_Q + (\lambda_A - c_Q)/\mu_A$. Under the maintained condition $0 < \lambda_A - c_Q < \mu_A(\Lambda - c_Q)$, effective asking friction satisfies $c_Q < c_Q^{eff} < \Lambda$. For type H with $\kappa_H > c_Q^{eff}/k$, because $c_Q^{eff} < \Lambda$, Π_{κ_H} is strictly increasing in a for any $m < k$ (slope $\mu_A(\Lambda - c_Q^{eff})(k - m) > 0$). The leader's unconstrained optimum for type H therefore sets $a_H^B = 1$ and interior specification $m_H^B = c_Q^{eff}/\kappa_H < k$. For type L with $\kappa_L < c_Q^{eff}/k$, the partial derivative with respect to effort satisfies $\partial \Pi_{\kappa_L}/\partial m = \mu_A[\Lambda(1 - a) + ac_Q^{eff} - \kappa_L m] \geq \mu_A[c_Q^{eff} - \kappa_L k] > 0$ for all $m \in [0, k]$ and all $a \in [0, 1]$. Hence Π_{κ_L} is strictly increasing in m , and attains its unconstrained maximum on $[0, k] \times [0, 1]$ at $m_L^B = k$. At $m_L^B = k$, zero attributes remain unspecified ($k - m_L^B = 0$), so asking is payoff-irrelevant to both the user and the leader ($\Pi_{\kappa_L}(k, a) = \mu_A[V - \frac{\kappa_L}{2}k^2]$ for all a). Setting $a_L^B = 0$ is the unconstrained benchmark choice because it gives zero asking service, minimizing the temptation of type H to mimic type L .

Step 2: Upward temptation to mimic (IC_L violation). Evaluating type κ_L 's payoff from mimicking the high type's bundle $(m_H^B, 1)$ versus accepting its own bundle $(k, 0)$:

$$U(m_H^B, 1; \kappa_L) - U(k, 0; \kappa_L) = \left[V - c_Q(k - m_H^B) - \frac{\kappa_L}{2}(m_H^B)^2 \right] - \left[V - \frac{\kappa_L}{2}k^2 \right].$$

Factoring terms:

$$U(m_H^B, 1; \kappa_L) - U(k, 0; \kappa_L) = (k - m_H^B) \left[\frac{\kappa_L(k + m_H^B)}{2} - c_Q \right].$$

Because $m_H^B < k$, this difference is strictly positive if and only if $\kappa_L(k + m_H^B) > 2c_Q$. When this condition holds, type κ_L strictly prefers the high type's asking contract over completing all attributes itself, violating IC_L.

Step 3: Slackness of participation (IR_H). In classical adverse selection with monetary transfers (Baron–Myerson, Laffont–Tirole), the principal's payoff is linearly reduced by agent rent, driving the inefficient agent's IR to bind at reservation $\underline{U}$. Here, the principal's payoff Π_{κ_H} directly incorporates $\mu_A U(\cdot; \kappa_H)$. Depressing U_H directly penalizes the leader's own objective. Thus, the leader adjusts the bundle (m_H, a_H) only as far as necessary to restore IC_L, leaving IR_H strictly slack: $U(m_H^{SB}, a_H^{SB}; \kappa_H) > \underline{U}$.

Step 4: Global optimization along the IC_L boundary and low-cost saturation. When IC_L binds, any feasible allocation must satisfy $U(m_H, a_H; \kappa_L) \leq U(m_L, a_L; \kappa_L)$. First, consider candidate menus that fix type L 's bundle at the unconstrained boundary $(m_L, a_L) = (k, 0)$, which fixes type L 's utility at $U_L^0 \equiv U(k, 0; \kappa_L) = V - \frac{\kappa_L}{2}k^2$. To implement this direct mechanism when the follower chooses specification effort m , the assistant specifies the off-menu deterrence schedule:

$$a(m) = \begin{cases} a_H^{SB}, & \text{if } m = m_H^{SB}, \\ a_L^{SB} = 0, & \text{if } m = m_L^{SB} = k, \\ 0, & \text{if } m \notin \{m_L^{SB}, m_H^{SB}\}. \end{cases}$$

Any deviation to an off-menu choice $m \notin \{m_L^{SB}, m_H^{SB}\}$ receives zero clarification ($a = 0$) and is strictly dominated for both types. (We note that this discontinuous step penalty is the standard

theoretical off-path construction of the revelation principle; practical implementations would employ smooth or threshold policies, as examined in Section 9.)

Along the binding boundary $U(m_H, a_H; \kappa_L) = U_L^0$, utility is strictly linear in a_H :

$$V - \Lambda(k - m_H) + a_H s(k - m_H) - \frac{\kappa_L}{2} m_H^2 = U_L^0,$$

where $s \equiv \Lambda - c_Q > 0$. Because $s(k - m_H) > 0$ for all $m_H < k$, this uniquely determines the required asking rate a_H as a function of specification effort m_H :

$$a_H(m_H) = 1 - \frac{U(m_H, 1; \kappa_L) - U_L^0}{s(k - m_H)}.$$

Next, evaluate the leader's payoff $\Pi_{\kappa_H}(m_H, a_H) = \mu_A U(m_H, a_H; \kappa_H) - b a_H(k - m_H)$ along this boundary, where $b \equiv \lambda_A - c_Q > 0$. Using the rent identity $U(m_H, a_H; \kappa_H) = U_L^0 - \frac{\kappa_H - \kappa_L}{2} m_H^2$ and substituting $a_H(m_H)(k - m_H)$, the leader's reduced objective becomes:

$$\Pi_{\kappa_H}(m_H) = \left(\mu_A - \frac{b}{s} \right) U_L^0 + \frac{b}{s} (V - \Lambda k) + \frac{b \Lambda}{s} m_H - \frac{1}{2} \left[\mu_A (\kappa_H - \kappa_L) + \frac{b}{s} \kappa_L \right] m_H^2.$$

The second derivative with respect to m_H is:

$$\frac{d^2 \Pi_{\kappa_H}}{d m_H^2} = - \left[\mu_A (\kappa_H - \kappa_L) + \frac{b}{s} \kappa_L \right] = - \frac{\mu_A s (\kappa_H - \kappa_L) + b \kappa_L}{s} < 0,$$

which is strictly negative everywhere on $[0, k]$ because $\kappa_H > \kappa_L$, $\mu_A > 0$, $s > 0$, and $b > 0$. Therefore, along the binding IC_L boundary, the leader's objective is globally strictly concave in m_H , with unique interior maximizer:

$$m_H^* = \frac{b \Lambda}{\mu_A s (\kappa_H - \kappa_L) + b \kappa_L} = \frac{\Lambda \tilde{\rho}}{s (\kappa_H - \kappa_L) + \tilde{\rho} \kappa_L},$$

where $\tilde{\rho} \equiv b/\mu_A = (\lambda_A - c_Q)/\mu_A$. Evaluating the asking rate at this global boundary optimum:

$$a_H^{SB} = a_H(m_H^*) = 1 - \frac{U(m_H^*, 1; \kappa_L) - U_L^0}{s(k - m_H^*)}.$$

Because $s(k - m_H^*) > 0$ for all $m_H^* < k$, strict downward distortion $a_H^{SB} < 1$ holds if and only if $U(m_H^*, 1; \kappa_L) - U_L^0 > 0$, which is strictly positive if and only if:

$$k > k_{\text{crit}}(\kappa_L, \kappa_H, \Lambda, c_Q, \lambda_A, \mu_A) \equiv \frac{2c_Q}{\kappa_L} - m_H^*. \quad (1)$$

Whenever $k > k_{\text{crit}}$, $U(m_H^*, 1; \kappa_L) > U_L^0$ holds strictly, which guarantees $a_H^{SB} < 1$.

We now analytically determine when $(m_L^{SB}, a_L^{SB}) = (k, 0)$ is the global optimum. Consider departing from $m_L = k$ by offering type L clarification service ($a_L = 1$) with $m_L < k$. Since $\kappa_L > c_Q/k$, reducing m_L below k increases type L 's utility at marginal rate $\left. \frac{\partial U}{\partial m_L} \right|_{m_L=k} = c_Q - \kappa_L k < 0$, yielding a utility increase of $\kappa_L k - c_Q > 0$. With population shares f_L, f_H , increasing type L 's utility by +1 relaxes IC_L , allowing a_H to increase by $\frac{1}{s(k - m_H)}$, which increases the leader's payoff on type H at rate $f_H \frac{\mu_A s - b}{s} (\kappa_L k - c_Q)$. Conversely, granting type L clarification costs the leader

$f_L \mu_A (c_Q^{eff} - \kappa_L k)$ per unit of reduction in m_L . Departing from $(k, 0)$ is strictly unprofitable if and only if the marginal gain on type H is weakly less than the marginal cost on type L :

$$f_H \frac{\mu_A s - b}{s} (\kappa_L k - c_Q) \leq f_L \mu_A (c_Q^{eff} - \kappa_L k).$$

Dividing by $f_L \mu_A$ and substituting $\tilde{\rho} = b/\mu_A$ and $t \equiv \frac{f_H}{f_L} \frac{s - \tilde{\rho}}{s}$, this simplifies to:

$$t(\kappa_L k - c_Q) \leq c_Q^{eff} - \kappa_L k \iff \kappa_L \leq \bar{\kappa}_L \equiv \frac{c_Q^{eff} + t c_Q}{k(1+t)}.$$

When $f_L = f_H = 0.5$, $t = \frac{s - \tilde{\rho}}{s}$ and $\bar{\kappa}_L$ simplifies to $\frac{sc_Q^{eff} + (s - \tilde{\rho})c_Q}{k(2s - \tilde{\rho})}$. When $\kappa_L \leq \bar{\kappa}_L$, the leader strictly prefers to keep type L at the corner $(k, 0)$. In this regime, $(m_L^{SB}, a_L^{SB}) = (k, 0)$ is the exact global second-best optimum, and our closed form for (m_H^*, a_H^{SB}) is exact. When $\bar{\kappa}_L < \kappa_L < c_Q^{eff}/k$, the leader departs from $(k, 0)$ by setting $a_L^{SB} = 1$ and $m_L^{SB} < k$. Raising U_L relaxes IC_L , allowing a_H^{SB} to rise toward 1 while preserving incentive compatibility.

Step 5: Comparative statics across the cost spectrum. In the low-cost saturation regime ($\kappa_L \leq \bar{\kappa}_L$):

$$1 - a_H^{SB} = \frac{\frac{\kappa_L(k + m_H^*)}{2} - c_Q}{s}.$$

As κ_L increases toward $\bar{\kappa}_L$, type L 's utility from completing k attributes decreases, intensifying the temptation to mimic the asking bundle. To deter mimicry, the leader must distort a_H^{SB} progressively downward (lower a_H). In the boundary relaxation regime ($\bar{\kappa}_L < \kappa_L < c_Q^{eff}/k$), the leader relieves this pressure by granting clarification to type L , which raises U_L and moderates the downward distortion on type H , so a_H^{SB} rises back toward 1 as $\kappa_L \rightarrow c_Q^{eff}/k$. Finally, when friction reaches or exceeds task stakes ($c_Q^{eff} \geq \Lambda$, or $\lambda_A - c_Q \geq \mu_A(\Lambda - c_Q)$), the leader unconstrainedly prefers zero asking for all types ($\partial \Pi / \partial a \leq 0$). Clarification is uniformly shut down: $a_L = a_H = 0$ and $m_L = m_H = k$. Both types receive the identical bundle $(k, 0)$, so IC_L and IC_H hold with equality ($0 = 0$). This boundary co-activity reflects technological clarification shutdown due to prohibitive friction, whereas genuine adverse-selection screening distortion ($a_H^{SB} < a_H^B = 1$) operates strictly in the cost-effective asking regime $c_Q < c_Q^{eff} < \Lambda$. $\square$

Remark 7 (Active Constraint Regimes and Clarification Shutdown). *In classical transfer-based screening (such as Baron–Myerson), IC_L and IR_H typically bind while IC_H and IR_L remain slack. In our setting, solving the full four-variable constrained problem reveals three distinct operational regimes:*

1. Small bias regime ($c_Q^{eff} \approx c_Q$): *When asking friction is close to true cost, the temptation of type L to mimic the asking bundle is weak or absent ($k \leq k_{crit}$), yielding negligible or zero distortion ($a_H^{SB} \approx a_H^B = 1$) with constraints slack.*
2. Moderate bias regime ($c_Q < c_Q^{eff} < \Lambda$ with $k > k_{crit}$): *Type L faces a strict temptation to mimic, and IC_L binds alone while IR_H , IC_H , and IR_L remain strictly slack. The leader directly penalizes itself by depressing follower utility, which keeps participation constraints slack and forces downward distortion $a_H^{SB} < a_H^B = 1$ purely to satisfy IC_L .*

3. Prohibitive friction regime ($c_Q^{eff} \geq \Lambda$, equivalently $\lambda_A - c_Q \geq \mu_A(\Lambda - c_Q)$): *Asking friction exceeds the entire task risk resolved, so the leader unconstrainedly prefers zero asking for all types ($\partial\Pi/\partial a \leq 0$). Clarification is uniformly shut down: $a_L = a_H = 0$ and $m_L = m_H = k$. Both types receive the identical bundle $(k, 0)$, so IC_L and IC_H hold with equality trivially. This boundary co-activity reflects technological shutdown, not an adverse-selection screening distortion.*

Remark 8 (Directionality of Policy Misalignment). *The downward distortion in Proposition 7 is specific to the under-asking bias case ($0 < \lambda_A - c_Q < \mu_A(\Lambda - c_Q)$), which reflects the empirical reality of turn-penalizing satisfaction metrics. Under an over-asking bias ($\lambda_A < c_Q$), the direction of distortion can reverse, as the leader would favor excessive clarification.*

7 Comparing the Two Regimes

Corollary 5 (Menus weakly dominate pooling). *The leader’s optimal expected payoff under Model II is weakly greater than under Model I: any pooling policy a is a feasible (degenerate) menu with $a_L = a_H = a$, trivially satisfying IC , so the unconstrained menu optimum is at least as good.*

Beyond payoff dominance, the central comparison lies in the structural decomposition of welfare losses across the two regimes, enabled by Propositions 4–7:

- **Model I (pooling):** Under the linear-risk specification on the physically feasible domain $m \in [0, k]$, an unbiased assistant selects $a^{SE} = 1$, which implements the first-best policy $a^{FB}(\kappa) = 1$ for every type in the population (Proposition 3). Consequently, in the absence of bias ($\lambda_A = c_Q$), the pooling restriction incurs zero welfare loss ($\Pi_{II} - \Pi_I = 0$). Welfare loss under pooling arises when objective bias is present ($\lambda_A > c_Q$), which suppresses the single ask rate a^{SE} below the socially efficient level (Corollary 2), or in the exact conjunctive model (Section 10), where first-best policies differ across types ($a^{FB} \in \{0, 1\}$) and a uniform rate cannot simultaneously tailor to heterogeneous needs.
- **Model II (screening)** incurs welfare loss from bias *alone* (Proposition 6): in the absence of bias, screening recovers full efficiency despite private cost information. Under bias, the loss takes the form of the downward distortion characterized in Proposition 7. This represents a fundamentally different mechanism from the instrument-restriction loss of pooling: it stems not from an inability to condition on observed effort, but from the requirement to prevent opportunistic mimicking across types.

7.1 Sensitivity of the Unbiased Pooling Corner and Physical Feasibility

While the qualitative finding of Corollary 4, that unbiased pooling is always a corner solution rather than an interior compromise, holds universally across the parameter space, the unclipped corner difference formula $\Pi(1) - \Pi(0) = (\Lambda - c_Q)[k - k^*]$ relies on unconstrained best responses. Table 1 documents this unconstrained boundary across varying task sizes k and baseline guessing accuracies g , holding $L = 10$, $c_Q = 2$, and $\kappa \sim \text{Uniform}[0.2, 0.6]$ (where $\mathbb{E}[1/\kappa] \approx 2.747$).

As shown in Table 1, once followers are restricted to feasible specification effort $m \in [0, k]$, asking ($a^{SE} = 1$) weakly dominates guessing across all 11 configurations under linear risk. The apparent dominance of guessing in the unclipped column stems entirely from low-cost types choosing unfeasible specification levels $m = \Lambda/\kappa > k$. At $k = 5$, for all $\kappa \in [0.2, 0.6]$, the unassisted specification choice under guessing is $\min\{5, \Lambda/\kappa\} = 5 = k$ across all $g \in \{0.30, 0.50, 0.70\}$, because

g	$\Lambda = L(1 - g)$	Threshold k^*	Task Size k	Unclipped $\Delta\Pi$	Clipped $\Delta\Pi$
0.30	7.0	12.36	5	-36.80	+0.15
0.30	7.0	12.36	10	-11.80	+5.49
0.30	7.0	12.36	15	+13.20	+20.10
0.50	5.0	9.61	5	-13.84	+0.15
0.50	5.0	9.61	9	-1.84	+3.69
0.50	5.0	9.61	10 (Benchmark)	+1.16	+5.42
0.50	5.0	9.61	12	+7.16	+9.62
0.50	5.0	9.61	15	+16.16	+17.12
0.70	3.0	6.87	5	-1.87	+0.15
0.70	3.0	6.87	8	+1.13	+1.71
0.70	3.0	6.87	10	+3.13	+3.32

Table 1: Sensitivity of the unbiased pooling corner choice across task complexity k and guessing capability g , for $L = 10, c_Q = 2, \kappa \sim \text{Uniform}[0.2, 0.6]$, where $\Delta\Pi \equiv \Pi(1) - \Pi(0)$. Columns compare the unconstrained closed form $(\Lambda - c_Q)[k - k^*]$ against the physically feasible clipped payoff difference computed via vectorized numerical integration (`underspec_sim/verifications/generate_table1.py`, recorded in `outputs/table1_unbiased_sensitivity.csv`). In the unclipped algebra, guessing appears to win whenever $k < k^*$ (e.g., -36.80 at $g = 0.30, k = 5$), but when followers are constrained to feasible effort $m \leq k$, asking strictly dominates across all rows (+0.15 at $g = 0.30, k = 5$), confirming Proposition 3.

$\Lambda \geq 3.0$ and $\kappa \leq 0.6$. When followers fully saturate specification effort under silence, the task risk parameter Λ drops out completely from the comparison, and the net gain from asking $\Delta\Pi = \mathbb{E}[U(m^*(1); 1) - U(k; 0)] = +0.15$ depends strictly on $c_Q = 2$ and κ . In contrast, when evaluated under the exact conjunctive success function $q(a, g)^{k-m}$, non-linear probability curvature generates genuine interior trade-offs where guessing beats asking even on the interior domain (Section 10).

Remark 9 (Welfare vs Leader Payoff Ordering Across Bias Bands). *While screening weakly dominates pooling in leader payoff ($\Pi_{II} \geq \Pi_I$) by revealed preference (Corollary 5), the ranking of user welfare $W \equiv \mathbb{E}[U]$ depends non-monotonically on provider asking friction λ_A (verified in `underspec_sim/verifications/verify_welfare_ordering.py`):*

1. Mild bias ($\lambda_A \in [2.5, 3.0]$): *Pooling maintains universal clarification ($a^{SE} = 1$) and achieves strictly higher user welfare than screening ($W_{\text{pool}} = 85.33$ vs $W_{\text{screen}} = 85.00$ at $\lambda_A = 2.5$, gap -0.33; $W_{\text{pool}} = 85.33$ vs $W_{\text{screen}} = 84.52$ at $\lambda_A = 3.0$, gap -0.81). Here, pooling forces the assistant to provide first-best clarification to everyone, whereas screening distorts asking downward on terse users ($a_H^{SB} < 1$) to protect provider payoff against information rents.*
2. Moderate bias ($\lambda_A \in [3.5, 4.5]$): *Pooling collapses to zero asking ($a^{SE} = 0$, yielding baseline unassisted welfare $W_{\text{pool}} = 80.00$), whereas screening continues to provide partial clarification to terse users ($a_H^{SB} > 0$), delivering strictly higher user welfare ($W_{\text{screen}} = 82.45$ at $\lambda_A = 3.5$, gap +2.45; $W_{\text{screen}} = 81.53$ at $\lambda_A = 4.0$, gap +1.53; $W_{\text{screen}} = 80.71$ at $\lambda_A = 4.5$, gap +0.71).*
3. Severe / prohibitive friction ($\lambda_A \geq 5.0$): *Clarification is priced out under both regimes ($a^{SE} = 0$ and $a_L = a_H = 0$), so user welfare ties at $W_{\text{pool}} = W_{\text{screen}} = 80.00$ (gap 0.00).*

In the exact conjunctive model, an additional structural instrument-restriction gap appears even at zero bias, separating technological restriction from behavioral misalignment.

8 Testable Predictions

- P1** (Prop. 1, Cor. 1) Holding the assistant’s policy fixed, higher guessing accuracy is associated with shorter specifications alongside weakly higher user welfare, and never with an overall welfare decline despite the reduction in prompt length.
- P1’** (Model I remark) Holding accuracy fixed, the effect of a higher *ask rate* on specification length flips sign depending on whether $c_Q \gtrless \Lambda = L(1 - g)$: where clarifying questions are inexpensive relative to the risk they resolve, users specify less as the ask rate rises (substitution); where questions are relatively costly to answer, users specify *more* (pre-emption). This prediction operates within the model and is distinct from P1’s accuracy channel.
- P2** (Prop. 2) The correlation between turn-level satisfaction and eventual task success weakens or inverts specifically for under-asking assistants, and can be benchmarked against the derived a^{SE} of Proposition 4 rather than an assumed ask rate.
- P3** Making ask-rate statistics visible to users shifts specification length toward the sophisticated best response analyzed in Section 3.3.
- P4** (Prop. 7) Ask rates should rise with a user’s revealed cost of providing detail (terseness) when clarification policies are screening menus or under nonlinear success models with attribute heterogeneity, rather than responding exclusively to model confidence.
- P5** Users with lower self-reported error-catching rates exhibit declining specification effort across an interactive session (following Section 3’s dynamic channel).
- P6** (Cor. 4, Prop. 7) Under an assistant with asking friction bias ($\lambda_A > c_Q$), a menu-screening policy achieves weakly higher leader payoff than pooling ($\Pi_{II} \geq \Pi_I$) by tailoring specification requirements to user capabilities. For user welfare ($W \equiv \mathbb{E}[U]$), however, the ranking depends non-monotonically on friction: under mild bias ($\lambda_A \in (c_Q, \lambda_A^*)$), pooling maintains universal clarification ($a^{SE} = 1$) and delivers higher user welfare than screening (which distorts $a_H^{SB} < 1$ downward to economize on leader friction); under severe bias ($\lambda_A \geq \lambda_A^*$), friction forces pooling to collapse to zero asking ($a^{SE} = 0$), whereas screening preserves partial clarification for high-cost users and restores higher user welfare. In settings characterized by nonlinear exact success or heterogeneous attribute stakes (where first best prescribes asking only for terse users), an unbiased pooling policy also incurs an instrument-restriction loss by forcing uniform asking.
- P7** (Prop. 7) Even when an assistant can condition on declared specification effort, terse users receive fewer clarifying questions than the assistant’s unconstrained per-type preference would prescribe ($a_H^{SB} < a_H^B$). This downward distortion arises specifically from the incentive-compatibility constraint IC_L . In the low-cost saturation regime ($\kappa_L \leq \bar{\kappa}_L$), downward distortion intensifies as cost heterogeneity narrows (κ_L approaches $\bar{\kappa}_L$), requiring deeper distortion to prevent mimicry; in the boundary relaxation regime ($\bar{\kappa}_L < \kappa_L < c_Q^{eff}/k$), the assistant grants clarification to low-cost users ($a_L^{SB} = 1$), relaxing IC_L and dampening downward distortion (a_H^{SB} rises toward 1 as $\kappa_L \rightarrow c_Q^{eff}/k$).
- P8** (Prop. 5) Under an assistant with under-asking friction bias ($\lambda_A > c_Q$), higher model guessing capability g perversely lengthens user prompts ($d\mathbb{E}[m^*]/dg > 0$) and lowers user welfare ($dW/dg < 0$). This equilibrium capability reversal operates because improved guessing reduces the assistant’s incentive to ask, forcing users to over-specify upstream to defensively insure against guessing errors.

9 Empirical Strategy and Experimental Design

The behavioral predictions P1–P8 (including P1′) translate our contract-theoretic mechanisms into observable margins for experimental testing. In naturalistic software engineering logs, however, assistant policy bias λ_A and user specification costs κ are typically confounded with task difficulty and user experience. To causally test our predictions, we propose a between-subjects randomized controlled trial on software developers performing standardized implementation tasks:

Experimental design and treatments. Participants are randomly assigned to interact with code assistants operating under distinct clarification regimes:

1. *Treatment A (Low-asking policy, $a = 0.1$):* The assistant aggressively guesses ambiguous attributes with minimal clarification.
2. *Treatment B (High-asking policy, $a = 0.8$):* The assistant frequently asks targeted clarifying questions before generating code.
3. *Treatment C (Menu-conditioned screening policy):* The assistant dynamically selects its clarification rate based on the structural completeness of the initial user prompt, implementing the screening menu derived in Section 6.
4. *Control (Unbiased benchmark):* The assistant’s clarification rate is calibrated to the user-optimal benchmark.

Measurement and identification. For each programming session, instrumentation logs: (i) upstream prompt length and specified invariants/edge cases (m); (ii) clarification turn count ($a(k - m)$); (iii) downstream test suite execution and correctness; and (iv) post-session subjective developer satisfaction. By exogenously varying the clarification policy a , this protocol directly tests whether users economize on prompt specification in response to anticipated assistance, distinguishes the substitution channel ($\Lambda > c_Q$) from the pre-emption channel ($c_Q > \Lambda$, Prediction P1′), and isolates the welfare losses attributable to objective bias versus pooling restrictions (Predictions P6, P7, and P8). To validate these protocols prior to live human deployment, we provide a companion open-source simulation harness (`underspec_sim/experiments_llm`) supporting both deterministic offline dry-run evaluation and live API interactions across pooling and screening treatments.

10 Discussion and Limitations

The linear-risk formulation is the main tractability baseline analyzed in Sections 4–7. Rather than viewing linear risk solely as a small-risk Taylor approximation, it is properly understood as an independent, additive-defect primitive model:

$$U(m, a; \kappa) = V - L(1 - q)(k - m) - \frac{\kappa}{2}m^2 - a c_Q (k - m),$$

where L represents the localized defect cost per unresolved attribute and cumulative defect loss satisfies $kL \leq V$. In this additive environment, each unaddressed attribute produces an independent defect without destroying the entire project. By contrast, the exact conjunctive specification $Vq(a, g)^{k-m}$ represents a systemic bottleneck environment where all attributes are strictly complementary and a single error destroys the entire deliverable. To calibrate the exact model against the linear baseline, note that for benchmark task size $k = 10$, total project valuation $V = 100$

corresponds precisely to cumulative defect loss $kL = 100$ (with per-attribute stake $L = 10$). To investigate the role of payoff curvature, we evaluate both the unbiased pooling optimum and the biased screening menu directly under the exact conjunctive payoff $q(a, g)^{k-m}$ via direct numerical grid search and constrained optimization (implemented in `verify_exact_conjunctive.py`, recorded in `outputs/exact_conjunctive_robustness.csv`). Rather than merely replicating the linear model, the exact conjunctive specification represents a distinct, richer economic environment where probability curvature introduces genuine interior trade-offs.

Under this exact nonlinear specification across task complexity $k \in [3, 20]$ and guessing accuracy $g \in [0.50, 0.95]$ (30 grid points):

1. The corner-solution property of Corollary 4 persists universally ($a_{\text{exact}}^{SE} \in \{0, 1\}$ in all 30 configurations).
2. Unbiased pooling choices agree between the exact and linear models in 17 of 30 cases (57%). The discrepancies reflect genuine nonlinear curvature that can make guessing optimal for moderate k and high guessing accuracy g . This curvature creates a positive instrument-restriction loss at zero bias in the exact model that is absent in linear risk.
3. Downward screening distortion ($a_H^{SB} < a_H^B$) holds in 20 of 30 cases (67% overall: 20% at $k \in \{3, 5\}$, 80% at $k \in \{8, 20\}$, and 100% at $k \in \{10, 15\}$). The low incidence of distortion at small k stems from boundary saturation: when tasks are simple ($k \leq 5$), users can fully specify all attributes ($m = k$), leaving no residual ambiguity for clarification to resolve ($k - m = 0$). Consequently, both IC_L and IC_H are active in 15 of 30 cases, concentrated entirely at small task sizes ($k \leq 8$).

Furthermore, numerical solutions of the full four-variable constrained menu problem under the exact payoff across $\lambda_A \in [c_Q, 10c_Q]$ and $\mu_A \in [0.1, 1.0]$ (`verify_exact_bias_sweep.py`, recorded in `outputs/exact_bias_sweep.csv` and Figure `outputs/exact_bias_sweep.png`) illustrate the boundary transition to clarification shutdown. In the linear model with $\mu_A = 1.0$, asking friction reaches or exceeds task stakes ($c_Q^{eff} \geq \Lambda$) at $\lambda_A \geq 5.0$. Beyond this threshold, the leader unconstrainedly prefers zero asking for all types ($a_L^B = a_H^B = 0$). Under the exact nonlinear payoff, this technological clarification shutdown threshold shifts outward to $\lambda_A \approx 8.0$. In this shutdown regime, both types choose full specification $m = k$ and receive $(k, 0)$, so IC_H and IC_L both hold with equality ($0 = 0$). This co-activity is not an adverse-selection screening distortion, but a technological shutdown artifact where clarification is universally priced out by severe asking friction.

Another important limitation concerns the off-menu contract schedule. Proposition 7 utilizes the standard direct revelation principle, implementing the second-best menu via a discontinuous off-menu policy $a(m) = 0$ for all $m \neq m_H^*$. While theoretically canonical, implementing discontinuous punishment schedules in conversational AI systems may be practically brittle or unintuitive for end users. Exploring smooth, continuous UI policies $a(m)$ or graded clarification schedules that remain robust to prompt gaming offers a fruitful avenue for applied mechanism design.

Regarding the regularity condition (Assumption 1), monotonicity of prompt specification ($m^*(g)$ weakly decreasing in g) holds whenever $(k - m) \ln q \geq -1$, which is guaranteed on the primitive domain whenever $k \ln q \geq -1$. While monotonicity can fail when the cross-partial turns strictly positive, empirical monotonicity can still survive locally in low-stakes regimes as a boundary property (`verify_monotonicity_survival.py`, recorded in `outputs/monotonicity_survival.csv`), because Topkis submodularity is a sufficient rather than necessary condition for comparative statics.

A stylized modeling assumption is attribute symmetry (k binary attributes with identical loss L and guessing probability g). In practical software engineering, attributes exhibit substantial

loss heterogeneity: omitting an architectural invariant or security convention causes severe failure ($L_i \gg 0$), while omitting a stylistic naming convention carries negligible cost ($L_j \approx 0$). In an extension with attribute-specific losses L_i , the user’s objective becomes $U = V \prod_i q_i - c(m) - \sum_i a_i c_Q$. Ordering attributes by descending risk stake $\Lambda_i = L_i(1 - g_i)$, users rationally specify the highest-stakes attributes first, while assistants prioritize asking on the highest-residual-stake attributes. The fundamental contract-theoretic mechanism, namely the substitution between upstream specification effort and automated clarification under misaligned asking friction, generalizes directly to the heterogeneous-loss setting.

A related limitation concerns the empirical identification of λ_A . Every bias-driven prediction (Corollary 2, Proposition 7, and predictions P2, P6, P7, and P8) is stated in terms of under-asking bias ($\lambda_A > c_Q$), but backing out λ_A from observed policy alone is circular. Lemma 1 formalizes this as an exact non-identification result: a^{SE} depends on (μ_A, λ_A) only through their ratio $\tilde{\rho} = (\lambda_A - c_Q)/\mu_A$, so a discounting-driven leader and a friction-driven leader with the identical excess friction ratio cannot be distinguished from pooling behavior alone. An independent empirical handle, such as an explicit penalty on turn count in the training objective or reward specification, would be needed before the bias-versus-heterogeneity predictions in Section 8 can be tested without circularity. The two-type restriction in Model II avoids ironing complications at the cost of generality; a continuum formulation using the standard Mirrlees technique (Mirrlees, 1971; Myerson, 1979) is a natural next step to characterize the continuous distribution of downward distortions across arbitrary cost distributions. We also assume truthful revelation of specified attributes (abstracting from cheap-talk ambiguity within what is communicated) and a single leader. Modeling competition among multiple assistants represents an appealing extension, as market competition could plausibly discipline asking bias directly.

A broader methodological conclusion concerns the limitations of applying classical contract-theoretic intuition directly to human–AI settings without accounting for non-monetary, partially aligned incentives. In classical mechanism design (such as Baron–Myerson, Laffont–Tirole, or costly verification models (Baron and Myerson, 1982; Laffont and Tirole, 1986; Ben-Porath et al., 2014)), the principal pays monetary transfers to extract surplus, and agent informational rent directly reduces the principal’s payoff. In human–AI interaction, the assistant’s objective directly incorporates user welfare. As a consequence, incentive compatibility is costless in the absence of bias (Proposition 6), participation constraints do not bind at the bottom of the type distribution, and unbiased pooling naturally induces corner solutions rather than interior compromises (Corollary 4). Economic analyses of automated assistance must therefore explicitly incorporate the non-monetary, partially aligned nature of the interaction rather than relying on direct analogies to cash-transfer mechanisms.

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